

Math 50 Midterm 1 Study Notes

Unit 1 (probability models) and Unit 2 (expectation, Normal, regression)

How to use these notes

Sections 1–2 are the concepts, in the same order as the “What you need to know” list on the exam page. The last page is a one-page landscape formula sheet. The 21 practice problems and their solutions are in the separate file `practice_solutions.pdf`. Everything here uses only the notation and tools from the course notes (in particular, correlation is ρ).

0. Must-memorize checklist

Distribution	Sample space / density	Mean	Variance
$Y \sim \text{Bernoulli}(q)$	$P(1) = q, P(0) = 1 - q$	q	$q(1 - q)$
$Y \sim \text{Uniform}(a, b)$	$f(y) = \frac{1}{b - a}$ on $[a, b]$	$\frac{a + b}{2}$	$\frac{(b - a)^2}{12}$ *
$Y \sim \text{Normal}(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/2\sigma^2}$	μ	σ^2

* The variance of a Uniform is *not* derived in the notes (that would need an integral), but the exam page says to memorize Uniform, and some practice problems need it. Bell curve: 68% within 1σ , 95% within 2σ , 99.7% within 3σ .

Also know cold: **(i)** $\mathbb{P}(x, y) = \mathbb{P}(y | x)\mathbb{P}(x)$ and Bayes; **(ii)** independence $\iff \mathbb{P}(x, y) = \mathbb{P}(x)\mathbb{P}(y)$; **(iii)** $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y | X]]$ (tower property, §2.2); **(iv)** $\text{var}(Y) = \mathbb{E}[\text{var}(Y | X)] + \text{var}(\mathbb{E}[Y | X])$ (law of total variance, §2.3); **(v)** $aX + b$ and sums of independent Normals; **(vi)** $\beta_1 = \mathbb{E}[Y | X = 1] - \mathbb{E}[Y | X = 0]$; **(vii)** $\text{cov}(X, Y) = \beta_1 \sigma_X^2$.

1 Unit 1: Probability models

1.1 Sample spaces, outcomes, events

A random variable is a quantity we cannot predict before observing it (capital letters X, Y, Z). Its **sample space** S (or S_X) is the set of all values it can take. Elements of S are **outcomes**; subsets of S are **events**. For an event U : $\mathbb{P}(U) = \sum_{x \in U} \mathbb{P}(x)$.

Axioms

Nonnegativity $\mathbb{P}(U) \geq 0$; normalization $\mathbb{P}(S) = 1$; additivity for disjoint events $\mathbb{P}(\cup U_i) = \sum \mathbb{P}(U_i)$.

Examples. Coin: $S = \{\text{heads}, \text{tails}\}$ (coded $\{1, 0\}$). Die: $S = \{1, \dots, 6\}$. Two independent coin-type variables (X_1, X_2) : $S = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$. Compound variables (Drill 4): list every reachable value; a value can be reached in more than one way (e.g. “die +1 if heads, die +4 if tails” gives 5 from $(H, 4)$ and $(T, 1)$).

1.2 The $\sim$ notation and the Bernoulli distribution

“ $Y \sim \text{Bernoulli}(q)$ ” means $Y \in \{0, 1\}$ with $\mathbb{P}(Y = 1) = q$, $\mathbb{P}(Y = 0) = 1 - q$. The probability of any event E is the probability of the Bernoulli variable $\mathbf{1}_E$. General pattern: *Variable* $\sim$ *Distribution(parameters)*.

1.3 Joint, marginal, conditional; independence

- **Marginal** (sum out the other variable): $\mathbb{P}(x) = \sum_y \mathbb{P}(x, y)$. On a table, add across a row or down a column.
- **Conditional**: $\mathbb{P}(x | y) = \frac{\mathbb{P}(x, y)}{\mathbb{P}(y)}$ (restrict to the y row/column, renormalize).
- **Product form**: $\mathbb{P}(x, y) = \mathbb{P}(x | y)\mathbb{P}(y)$. **Bayes**: $\mathbb{P}(x | y) = \frac{\mathbb{P}(y | x)\mathbb{P}(x)}{\mathbb{P}(y)}$.
- Conditioning is *not* symmetric: $\mathbb{P}(A | B) \neq \mathbb{P}(B | A)$ in general.
- **Independent** $\iff \mathbb{P}(x, y) = \mathbb{P}(x)\mathbb{P}(y)$ for all $x, y \iff \mathbb{P}(y | x) = \mathbb{P}(y) \iff \mathbb{P}(x | y) = \mathbb{P}(x)$.

Exam move: “are X and Y independent? Justify with a single calculation”

To show *not* independent you only need *one* cell where $\mathbb{P}(x, y) \neq \mathbb{P}(x)\mathbb{P}(y)$ (or one conditional that differs from its marginal). To show independent you must check every cell. Be careful when the numbers are close: compute exactly, do not eyeball.

Worked example (from the notes). $\mathbb{P}(Y_A, Y_B)$: $(0, 0): \frac{1}{2}$, $(0, 1): \frac{1}{8}$, $(1, 0): \frac{1}{8}$, $(1, 1): \frac{1}{4}$. Marginals: $\mathbb{P}(Y_A = 0) = \frac{5}{8}$, $\mathbb{P}(Y_B = 0) = \frac{5}{8}$. Then $\mathbb{P}(Y_A = 0)\mathbb{P}(Y_B = 0) = \frac{25}{64} \neq \frac{1}{2}$, so not independent. Conditional: $\mathbb{P}(Y_A = 1 | Y_B = 0) = \frac{1/8}{5/8} = \frac{1}{5}$.

1.4 From a $\sim$ model to the joint table, and back

Model $\rightarrow$ table. A model like

$$Y \sim \text{Bernoulli}(1/2), \quad X | Y \sim \text{Bernoulli}(Y/4 + 1/4)$$

means: first draw Y ; then, given Y , draw X with success probability $Y/4 + 1/4$. So $\mathbb{P}(X=1 | Y=0) = \frac{1}{4}$, $\mathbb{P}(X=1 | Y=1) = \frac{1}{2}$. Multiply along branches: $\mathbb{P}(x, y) = \mathbb{P}(x | y)\mathbb{P}(y)$.

$\mathbb{P}(x, y)$	$Y = 0$	$Y = 1$	
$X = 0$	$\frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}$	$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$	
$X = 1$	$\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$	$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$	
	$\frac{1}{2}$	$\frac{1}{2}$	1

Table → model. Read Y 's marginal off the column sums, then divide each column by its sum to get $X | Y$, and recognize the conditional as a named distribution whose parameter depends on y . Always check the table sums to 1.

1.5 Continuous distributions

A continuous random variable has a **density** $f(y) \geq 0$ with $\int f = 1$, and probabilities are areas:

$$\mathbb{P}(a < Y < b) = \int_a^b f(y) dy, \quad \text{CDF: } F(y) = \mathbb{P}(Y \leq y).$$

- $\mathbb{P}(Y = y) = 0$ for every single point; only intervals have positive probability.
- **A density is not a probability:** $f(y)$ can exceed 1 (e.g. $\text{Uniform}(0, 0.5)$ has $f = 2$); only $\int f \leq 1$. Think of $f(y)\Delta y \approx \mathbb{P}(y \leq Y \leq y + \Delta y)$.
- $Y \sim \text{Uniform}(a, b)$: $f(y) = \frac{1}{b-a}$ on $[a, b]$, so $\mathbb{P}(y_1 < Y < y_2) = \frac{y_2 - y_1}{b-a}$ (proportional to length).
- Conditioning a Uniform on a sub-interval gives a Uniform on the sub-interval: $A \sim \text{Uniform}(20, 90)$, $A | (A < 50) \sim \text{Uniform}(20, 50)$. Equivalent recipe: $\mathbb{P}(Y < 1 | Y < 3) = \mathbb{P}(Y < 1) / \mathbb{P}(Y < 3)$ because $\{Y < 1\} \subset \{Y < 3\}$.
- No integrals will be required; Normal probabilities use the bell-curve rule (Unit 2).

1.6 Sampling and simulation: translating code to probability statements

A simulation approximates probability by frequency: $\mathbb{P}(X = x) \approx N(x)/n$. **Key idea:** a probability is the mean of a boolean array, and conditioning means *restricting the array first*, then averaging.

Python	Probability meaning
<code>X = np.random.choice(vals, p=p, size=N)</code>	N iid draws with $\mathbb{P}(X = \text{vals}[i]) = p[i]$
<code>np.mean(X == x)</code>	$\mathbb{P}(X = x)$
<code>np.mean((X == x) & (Y == y))</code>	$\mathbb{P}(X = x, Y = y)$
<code>np.mean(X[Y == y] == x)</code>	$\mathbb{P}(X = x Y = y)$
<code>np.mean(X)</code>	$\mathbb{E}[X]$
<code>np.mean(X[Y == y])</code>	$\mathbb{E}[X Y = y]$
<code>np.var(X)</code>	$\text{var}(X)$
<code>np.random.normal(mu, sigma, n)</code>	draws from $\text{Normal}(\mu, \sigma^2)$ (takes the <i>standard deviation</i>)
<code>np.random.uniform(a, b, n)</code>	draws from $\text{Uniform}(a, b)$
<code>np.cov(x, y)[0,1] / np.cov(x, y)[0,0]</code>	estimates $\text{cov}(X, Y) / \text{var}(X) = \beta_1$

Reading a simulation as a model. An if/mask that draws x with different parameters depending on y is a conditional model $X | Y$:

```
y = np.random.choice([0, 1], p=[0.4, 0.6])    # Y ~ Bernoulli(0.6)
if y == 0:
```

```
x = np.random.choice([0, 1], p=[0.8, 0.2])    # X | Y=0 ~ Bernoulli(0.2)
else:
    x = np.random.choice([0, 1], p=[0.3, 0.7])    # X | Y=1 ~ Bernoulli(0.7)
```

Model: $Y \sim \text{Bernoulli}(0.6)$, $X | Y \sim \text{Bernoulli}(0.2 + 0.5Y)$. Joint: $\mathbb{P}(X=1, Y=0) = 0.4(0.2) = 0.08$, $\mathbb{P}(X=1, Y=1) = 0.6(0.7) = 0.42$, so $\mathbb{P}(X=1) = 0.50$. Independent? No, since $\mathbb{P}(X=1 | Y=0) = 0.2 \neq 0.7 = \mathbb{P}(X=1 | Y=1)$.

Vectorized mask pattern (used on the exam page's practice problems).

```
x = np.random.choice([0, 1], size=n, p=[0.3, 0.7])
y = np.empty(n)
y[x == 0] = np.random.uniform(0, 2, len(y[x == 0]))    # Y | X=0 ~ Uniform(0,2)
y[x == 1] = np.random.uniform(1, 4, len(y[x == 1]))    # Y | X=1 ~ Uniform(1,4)
```

Each line assigns a *conditional* distribution to the rows selected by the mask. Also, `np.random.normal(3 + 4*x, 2, n)` means $Y | X \sim \text{Normal}(3 + 4X, 2^2)$ (mean is an array, so it depends on each row's x ; the 2 is a standard deviation, so the variance is 4).

Predict-the-output rules of thumb.

1. Identify the model (in $\sim$ notation) first. Then the printed number is the corresponding exact quantity, up to sampling error ($\approx 1/\sqrt{n}$).
2. `np.mean(x)` $\rightarrow \mathbb{E}[X]$ computed by the tower property (weight conditional means by group probabilities).
3. `np.mean(x[y==1])` $\rightarrow \mathbb{E}[X | Y = 1]$: just the parameter used in that branch.
4. `np.mean(y[x>0])` $\rightarrow \mathbb{P}(Y = 1 | X > 0)$ when y is 0/1 (a mean of 0/1 values is a probability).
5. Two arrays generated in separate lines given the same x are *conditionally independent given x* , but generally not independent marginally.

2 Unit 2: Expectation, the Normal, and regression

2.1 Expectation, variance, CV

$$\mathbb{E}[Y] = \sum_y y \mathbb{P}(y), \quad \mathbb{E}[g(Y)] = \sum_y g(y) \mathbb{P}(y), \quad \text{var}(Y) = \mathbb{E}[(Y - \mathbb{E}Y)^2] = \mathbb{E}[Y^2] - (\mathbb{E}Y)^2.$$

Standard deviation $\sigma = \sqrt{\text{var}(Y)}$ (same units as Y). **Coefficient of variation** $\text{CV} = \sigma/\mu$ (unitless: compares spread across different scales; blows up when the mean is near 0; shifting all values by a constant changes μ but not σ).

Properties of expectation

1. Linearity: $\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$ (always, no independence needed). So $\mathbb{E}[aX+b] = a\mathbb{E}[X] + b$.
2. $\text{var}(aX+b) = a^2 \text{var}(X)$ (shifts don't matter, scale gets squared).
3. Independent $\Rightarrow \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, and $\text{var}(X+Y) = \text{var}(X) + \text{var}(Y)$.

Bernoulli/Binomial facts. $Y \sim \text{Bernoulli}(q)$: $\mathbb{E}Y = q$, $\text{var} Y = q(1-q)$ (since $Y^2 = Y$). If $Y = \sum_{i=1}^N X_i$ with X_i iid $\text{Bernoulli}(q)$: $\mathbb{E}Y = Nq$, $\text{var} Y = Nq(1-q)$, $\text{CV} = \sqrt{(1-q)/(qN)} \rightarrow 0$.

2.2 The tower property

Conditional expectation $\mathbb{E}[Y \mid X = x] = \sum_y y \mathbb{P}(y \mid x)$: from data, average Y over only those rows with $X = x$. $\mathbb{E}[Y \mid X]$ is itself a random variable (a function of X). Careful: $\mathbb{E}[Y \mid X = x] = \mathbb{E}[Y]$ for all x does *not* imply independence.

Tower property

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]] = \sum_x \mathbb{E}[Y \mid X = x] \mathbb{P}(X = x).$$

The overall average is a probability-weighted average of the group averages.

Worked example. Morning ($X=0$, prob 0.6, mean 78), afternoon ($X=1$, prob 0.4, mean 86):

$$\mathbb{E}[Y] = 78(0.6) + 86(0.4) = 81.2.$$

2.3 Law of total variance

Law of total variance

$$\text{var}(Y) = \underbrace{\mathbb{E}[\text{var}(Y \mid X)]}_{\text{within-group}} + \underbrace{\text{var}(\mathbb{E}[Y \mid X])}_{\text{between-group}}.$$

Total spread splits into the average spread *inside* each group, plus how much the group *means* themselves vary.

Worked example. Continuing the morning/afternoon example above, with both groups having variance 64:

$$\mathbb{E}[\text{var}(Y \mid X)] = 64,$$

$$\text{var}(\mathbb{E}[Y \mid X]) = 0.6(78 - 81.2)^2 + 0.4(86 - 81.2)^2 = 15.36,$$

so $\text{var}(Y) = 64 + 15.36 = 79.36$. Total variance is bigger than either group's, because the group means also differ.

2.4 The Normal distribution

$X \sim \text{Normal}(\mu, \sigma^2)$ is symmetric, bell-shaped, determined by mean μ (center) and variance σ^2 (spread). **Watch the convention:** math notation $\text{Normal}(\mu, \sigma^2)$ takes the variance; `np.random.normal(mu, sigma)` takes the standard deviation.

Standardizing. $Z = \frac{X - \mu}{\sigma} \sim \text{Normal}(0, 1)$: measures X in standard-deviation units.

Bell-curve rule.

Interval	$\mu \pm \sigma$	$\mu \pm 2\sigma$	$\mu \pm 3\sigma$
Probability	$\approx 68\%$	$\approx 95\%$	$\approx 99.7\%$

By symmetry, each tail beyond 1σ is $\approx 16\%$ ($(100 - 68)/2$), beyond 2σ is $\approx 2.5\%$, beyond 3σ is $\approx 0.15\%$. Also $\mathbb{P}(Z < 1) \approx 0.84$. **Recipe:** standardize the endpoints, then count how many σ 's they are from μ .

Linear transformations and sums

If $X \sim \text{Normal}(\mu_x, \sigma_x^2)$ then $aX + b \sim \text{Normal}(a\mu_x + b, a^2\sigma_x^2)$.

If $X_1 \sim \text{Normal}(\mu_1, \sigma_1^2)$, $X_2 \sim \text{Normal}(\mu_2, \sigma_2^2)$ **independent**:

$$aX_1 + bX_2 + d \sim \text{Normal}(a\mu_1 + b\mu_2 + d, a^2\sigma_1^2 + b^2\sigma_2^2).$$

Variances add (with squared coefficients) even for a difference: $\text{var}(X_1 - X_2) = \sigma_1^2 + \sigma_2^2$. Average of N independent copies of $\text{Normal}(\mu, \sigma^2)$ is $\text{Normal}(\mu, \sigma^2/N)$: same center, smaller spread.

Example. $Y \sim \text{Normal}(5, 4)$, so $\sigma = 2$. $\mathbb{P}(Y > 7)$: $7 = \mu + \sigma$, so $\approx 16\%$. $\mathbb{P}(Y > 3 \mid Y < 7) = \frac{\mathbb{P}(3 < Y < 7)}{\mathbb{P}(Y < 7)} \approx \frac{0.68}{0.84} \approx 0.81$.

2.5 The single-predictor regression model

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma^2) \iff Y = \beta_0 + \beta_1 X + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma^2).$$

- β_0 : intercept, $\mathbb{E}[Y \mid X = 0]$ (the line's height at $x = 0$).
- β_1 : slope, the change in average Y per one-unit increase in X .
- σ^2 ($= \sigma_\epsilon^2$): noise variance, the vertical spread of Y around the line (same at every x).

Assumptions.

1. *Linearity*: $\mathbb{E}[Y \mid X] = \beta_0 + \beta_1 X$ is a straight line.
2. *Homoskedasticity*: $\text{var}(Y \mid X) = \sigma^2$ does not depend on X .
3. *Conditional normality*: $Y \mid X$ (equivalently ϵ) is Normal.
4. *Exogeneity*: $\mathbb{E}[\epsilon \mid X] = 0$, i.e. nothing left in the noise is related to X . If it fails, X is *endogenous*.

Spotting failures on a scatter plot: curved cloud $\rightarrow$ linearity fails; spread that fans out or shrinks with $x \rightarrow$ homoskedasticity fails; strong skew or outliers in the vertical direction $\rightarrow$ normality fails.

Exogeneity cannot be checked from a scatter plot of (x, y) alone: it is a statement about unobserved factors inside ϵ , and the plot only shows how Y relates to X (the effect of X and the effect of any confounder look identical). Exogeneity is guaranteed by *randomized assignment* of X (coin flip), and threatened by self-selection.

2.6 Binary predictor: β_1 is a difference of means

If $X \in \{0, 1\}$ (control/treatment), then $\mathbb{E}[Y \mid X = 0] = \beta_0$ and $\mathbb{E}[Y \mid X = 1] = \beta_0 + \beta_1$, so

$$\boxed{\beta_1 = \mathbb{E}[Y \mid X = 1] - \mathbb{E}[Y \mid X = 0].}$$

Estimates: $\hat{\beta}_0 = \bar{Y}_{X=0}$, $\hat{\beta}_1 = \bar{Y}_{X=1} - \bar{Y}_{X=0}$ (in code, `y[x==1].mean() - y[x==0].mean()`). Relabeling ($X' = 1 - X$) swaps the groups: $\beta'_0 = \beta_0 + \beta_1$, $\beta'_1 = -\beta_1$. With a fair coin, $\mathbb{E}[Y] = \beta_0 + \beta_1/2$ (tower property).

2.7 Covariance and the slope

$\text{cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$, and $\text{cov}(X, X) = \text{var}(X)$.

Key derivation: $\text{cov}(X, Y) = \beta_1 \sigma_X^2$

Using $\mathbb{E}[XY] = \mathbb{E}[X \mathbb{E}[Y | X]] = \mathbb{E}[X(\beta_0 + \beta_1 X)] = \beta_0 \mathbb{E}[X] + \beta_1 \mathbb{E}[X^2]$ and $\mathbb{E}[X^2] = \sigma_X^2 + \mu_X^2$:

$$\text{cov}(X, Y) = \beta_0 \mu_X + \beta_1 (\sigma_X^2 + \mu_X^2) - \mu_X (\beta_0 + \beta_1 \mu_X) = \beta_1 \sigma_X^2.$$

Hence $\beta_1 = \text{cov}(X, Y) / \text{var}(X)$, valid for any X (not just binary), and $\text{var}(Y) = \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2$ (total variance: within = σ_ϵ^2 , between = $\beta_1^2 \sigma_X^2$).

Marginal of Y when X is Normal: $Y \sim \text{Normal}(\beta_0 + \beta_1 \mu_X, \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2)$ (sum of independent Normals). When X is *not* Normal (e.g. Bernoulli), Y still has the same mean and variance formulas, but Y is a mixture and is *not* Normal.

2.8 Correlation, R^2 , regression to the mean

$$\rho = \frac{\beta_1 \sigma_X}{\sigma_Y} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}, \quad \mathbb{E}[Z_Y | Z_X] = \rho Z_X, \quad \rho^2 = 1 - \frac{\sigma_\epsilon^2}{\sigma_Y^2} = R^2, \quad -1 \leq \rho \leq 1.$$

ρ is the regression slope after standardizing both variables (unitless); ρ^2 is the fraction of $\text{var}(Y)$ explained by X ($\rho^2 = 1$: Y is a deterministic line in X ; $\rho = 0$: X tells you nothing about the mean of Y). **Regression to the mean:** if $|\rho| < 1$, an unusually extreme X (say z SDs from its mean) predicts a *less* extreme Y (ρz SDs); this is not a causal effect, just luck not repeating. Rescaling Y (e.g. units) changes β_1 and cov but not ρ or R^2 .

2.9 Exogeneity and randomized experiments (conceptual)

Random assignment (a fair coin) makes X independent of everything else, so $\mathbb{E}[\epsilon | X] = 0$ and $\hat{\beta}_1$ can be read as the effect of the treatment. If subjects self-select into groups, unmeasured traits (motivation, health) that affect Y also differ between groups: $\mathbb{E}[\epsilon | X] \neq 0$, X is endogenous, and $\hat{\beta}_1$ mixes the treatment effect with those differences.

Named distributions

	pmf / density	mean	var
Bernoulli(q)	$\mathbb{P}(0) = 1 - q, \mathbb{P}(1) = q$	q	$q(1 - q)$
Uniform(a, b)	$\frac{1}{b-a}$ on $[a, b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal(μ, σ^2)	(below)	μ	σ^2
Normal density $\frac{1}{\sqrt{2\pi\sigma^2}}e^{-(x-\mu)^2/(2\sigma^2)}$; second parameter is the <i>variance</i> .			
Bell curve	$\mu \pm \sigma$	$\mu \pm 2\sigma$	$\mu \pm 3\sigma$
inside	68%	95%	99.7%
one tail beyond	16%	2.5%	0.15%
$\mathbb{P}(Z < 1) \approx 84\%, \mathbb{P}(Z < 0) = 50\%$.			

Probability

Events: $\mathbb{P}(U) = \sum_{x \in U} \mathbb{P}(x)$, $\mathbb{P}(S) = 1$,
 $\mathbb{P}(\text{not } U) = 1 - \mathbb{P}(U)$; “at least one” = $1 - \mathbb{P}(\text{none})$.
Marginal: $\mathbb{P}(x) = \sum_y \mathbb{P}(x, y)$; through a conditional:
 $\mathbb{P}(y) = \sum_x \mathbb{P}(y | x)\mathbb{P}(x)$.

Conditional ($\mathbb{P}(y) > 0$): $\mathbb{P}(x | y) = \frac{\mathbb{P}(x, y)}{\mathbb{P}(y)}$, so

$$\mathbb{P}(x, y) = \mathbb{P}(x | y)\mathbb{P}(y).$$

Bayes: $\mathbb{P}(x | y) = \frac{\mathbb{P}(y | x)\mathbb{P}(x)}{\mathbb{P}(y)}$; $\mathbb{P}(A | B) \neq \mathbb{P}(B | A)$.

X, Y **independent** $\iff \mathbb{P}(x, y) = \mathbb{P}(x)\mathbb{P}(y)$ for all x, y
 $\iff \mathbb{P}(y | x) = \mathbb{P}(y) \iff \mathbb{P}(x | y) = \mathbb{P}(x)$.

Continuous: density $f \geq 0$, $\int f = 1$; CDF

$$F(y) = \mathbb{P}(Y \leq y); \mathbb{P}(a < Y < b) = \int_a^b f = F(b) - F(a);$$

$\mathbb{P}(Y=y) = 0$; f can exceed 1. Sub-interval:

$$\mathbb{P}(Y < 1 | Y < 3) = \mathbb{P}(Y < 1) / \mathbb{P}(Y < 3).$$

 $\sim$ notation $\leftrightarrow$ joint table

$Y \sim \text{Bernoulli}(q)$, $X | Y \sim \text{Bernoulli}(g(Y))$:

$$\mathbb{P}(X=1, Y=y) = g(y)\mathbb{P}(y),$$

$$\mathbb{P}(X=0, Y=y) = (1 - g(y))\mathbb{P}(y), \quad \mathbb{P}(X=1) = \sum_y g(y)\mathbb{P}(y).$$

Table $\rightarrow$ model: sum over x to get $\mathbb{P}(y)$; divide each y -slice by $\mathbb{P}(y)$ to get $X | Y=y$. Check the table sums to 1.

Exam moves.

- Independence: one failing cell $\Rightarrow$ dependent.
- Simulation: write the model in $\sim$ form first; a printed mean $\approx$ the exact $\mathbb{E}$ (tower).
- Mean of a 0/1 array is a probability.
- Sample average of rows meeting a condition = conditional expectation.
- Densities can exceed 1; mixtures of Normals are not Normal.

Expectation and variance

$\mathbb{E}[g(Y)] = \sum_y g(y)\mathbb{P}(y)$; $\mathbb{E}[Y | x] = \sum_y y \mathbb{P}(y | x)$ (data: average over rows with $X=x$).

$$\text{var } Y = \mathbb{E}[Y^2] - (\mathbb{E}Y)^2, \quad \sigma = \sqrt{\text{var } Y};$$

$$\text{var}(Y | x) = \mathbb{E}[Y^2 | x] - \mathbb{E}[Y | x]^2.$$

$\mathbb{E}[Y^2] \neq (\mathbb{E}Y)^2$: $\mathbb{E}[Y^2] = \text{var } Y + (\mathbb{E}Y)^2 \geq (\mathbb{E}Y)^2$, equal only if $\text{var } Y = 0$. Compute $\mathbb{E}[Y^2] = \sum_y y^2 \mathbb{P}(y)$ (square first, then average). Bernoulli: $Y^2 = Y$, so $\mathbb{E}[Y^2] = q$.

CV = σ/μ : unitless; shifting values changes μ , not σ ;

blows up near $\mu = 0$.

$$\mathbb{E}[aX + b] = a\mathbb{E}X + b, \quad \text{var}(aX + b) = a^2 \text{var } X.$$

$\mathbb{E}[X + Y] = \mathbb{E}X + \mathbb{E}Y$ always. If independent:

$$\mathbb{E}[XY] = \mathbb{E}X \mathbb{E}Y \text{ and } \text{var}(X + Y) = \text{var } X + \text{var } Y.$$

Binomial (N iid Bernoulli(q)): $\mathbb{E} = Nq$, $\text{var} = Nq(1 - q)$,

$$\text{CV} = \sqrt{\frac{1-q}{qN}}.$$

Tower: $\mathbb{E}Y = \sum_x \mathbb{E}[Y | x] \mathbb{P}(x)$.

Total variance:

$$\text{var } Y = \mathbb{E}[\text{var}(Y | X)] + \text{var}(\mathbb{E}[Y | X])$$

(within-group + between-group)

Normal

Standardize: $Z = \frac{X - \mu}{\sigma} \sim \text{Normal}(0, 1)$. Recipe:

standardize the endpoints, count σ 's.

$$aX + b \sim \text{Normal}(a\mu + b, a^2\sigma^2).$$

Independent sum:

$$aX_1 + bX_2 + d \sim \text{Normal}(a\mu_1 + b\mu_2 + d, a^2\sigma_1^2 + b^2\sigma_2^2);$$
 a

difference still *adds* variances.

Mean of N iid: $\text{Normal}(\mu, \sigma^2/N)$.

$$\mathbb{P}(X > \mu) = \frac{1}{2} \text{ (symmetry)}.$$

A mixture of Normals with different means (e.g. $Y | X$ with X Bernoulli) is *not* Normal.

Regression model

$Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma^2) \iff Y = \beta_0 + \beta_1 X + \epsilon$,
 $\epsilon \sim \text{Normal}(0, \sigma^2)$.

$\beta_0 = \mathbb{E}[Y | X=0]$; β_1 = change in mean Y per unit X ;
 $\sigma^2 = \text{var}(Y | X)$ = vertical spread, same at every x .

Assumptions: (1) linear mean; (2) constant variance;

(3) Normal errors; (4) exogeneity $\mathbb{E}[\epsilon | X] = 0$.

Fails on a plot: curve $\rightarrow$ (1); fanning spread $\rightarrow$ (2);

skew/outliers $\rightarrow$ (3). (4) **cannot be checked from a**

scatter plot. Randomization $\Rightarrow$ exogenous; self-selection
 $\Rightarrow$ maybe endogenous.

Binary predictor $X \in \{0, 1\}$

$$\beta_1 = \mathbb{E}[Y | X=1] - \mathbb{E}[Y | X=0]$$

$$\hat{\beta}_0 = \bar{Y}_{X=0}, \quad \hat{\beta}_1 = \bar{Y}_{X=1} - \bar{Y}_{X=0}$$

$$\mathbb{E}Y = \beta_0 + \beta_1 \mathbb{P}(X=1)$$

$X \sim \text{Bernoulli}(p)$: $\sigma_X^2 = p(1 - p)$ ($\frac{1}{4}$ if fair), so

$$\text{cov} = \beta_1 p(1 - p).$$

Relabel $X' = 1 - X$: $\beta'_0 = \beta_0 + \beta_1$, $\beta'_1 = -\beta_1$.

Covariance, correlation, R^2

$$\text{cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}X \mathbb{E}Y, \quad \text{cov}(X, X) = \text{var } X$$

$$\text{cov}(X, Y) = \beta_1 \sigma_X^2 \Rightarrow \beta_1 = \frac{\text{cov}(X, Y)}{\text{var } X}$$

any X with $\text{var } X < \infty$

$\mathbb{E}Y = \beta_0 + \beta_1 \mathbb{E}X$; $\text{var } Y = \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2$ (between + within).

X Normal: $Y \sim \text{Normal}(\beta_0 + \beta_1 \mu_X, \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2)$. X not

Normal: same mean and variance, but Y is not Normal.

$$\rho = \frac{\beta_1 \sigma_X}{\sigma_Y} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1], \text{ sign of } \rho = \text{sign of } \beta_1.$$

$$R^2 = \rho^2 = 1 - \frac{\sigma_\epsilon^2}{\sigma_Y^2}; \quad \mathbb{E}[Z_Y | Z_X] = \rho Z_X \text{ (regression to the}$$

mean).

Changing units of Y changes β_1 , cov; not ρ , R^2 .

Independent $\Rightarrow$ cov = 0, but cov = 0 $\nRightarrow$ independent.

Code $\rightarrow$ math

<code>np.random.choice(vals, p=p, size=n)</code>	n iid draws from pmf p
<code>np.random.uniform(a, b, n)</code>	Uniform(a, b)
<code>np.random.normal(mu, sigma, n)</code>	Normal(μ, σ^2), SD input
<code>np.random.normal(3+4*x, 2, n)</code>	Normal($3 + 4X, 2^2$)
<code>y[x==0] = ...</code>	$Y X=0$
<code>np.mean(X==x)</code>	$\mathbb{P}(X=x)$
<code>np.mean((X==x) & (Y==y))</code>	$\mathbb{P}(X=x, Y=y)$
<code>np.mean(X[Y==y]==x)</code>	$\mathbb{P}(x y)$
<code>np.mean(X) / np.var(X)</code>	$\mathbb{E}X / \text{var } X$
<code>np.mean(X[Y==y])</code>	$\mathbb{E}[X y]$
<code>y[x==1].mean() - y[x==0].mean()</code>	$\approx \beta_1$ (binary x)
<code>np.cov(x, y)[0, 1] /</code>	$\approx \beta_1$
<code>np.cov(x, y)[0, 0]</code>	